From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST-III

PAPER -1

ANSWERS KEY			
	PHYSICS	CHEMISTRY	MATHEMATICS
Q. No.	ANSWER	ANSWER	ANSWER
1.	A	A	C
2.	D	C	C
3.	A	C	B
4.	A	A	A
5.	C	A	D
6.	A	D	C
7.	A	D	A
8.	A, B, C, D	A, C	A, C, D
9.	A, B, C	B, C	A, B
10.	A, B, C, D	A, B, D	A, B
11.	A, B, D	B, C, D	A, C
12.	B	D	C
13.	A	C	D
14.	A	B	D
15.	A	D	B
16.	D	D	D
1.	4	3	3
2	0	4	0
3	4	6	3
4	3	2	4
5	2	3	3
6	2	3	2
7	5	3	4

Physics

PART – I

SECTION – A

1. $\varepsilon_{avg} = \dfrac{\pi r^2 \Delta B}{\Delta t}$

$\therefore \quad \varepsilon_{avg} = \pi r^2 \Delta B \sqrt{\dfrac{g}{2h}}$

3. Magnitude of net force = Bi (Separation between ends) = 2BiR

5. At t = 0 capacitor behaves as closed circuit,

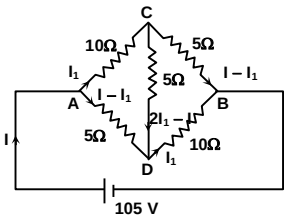
$10I_1 + 5(2I_1 - I) - 5(I - I_1) = 0$ . . . (1)

$2I_1 + 2I_1 - I - I + I_1 = 0$

$\Rightarrow \quad I = 5I_1/2$

$5(I - I_1) + 10I_1 = 105$

$\Rightarrow \quad I_1 = 6A$



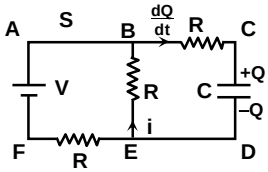
7. Energy is conserved.

14. Apply KVL in ABEFA and BCEDB to get

$Q = \dfrac{VC}{2}(1 - e^{-2t/3RC})$

15. To calculate current in ED find dQ/dt

$\dfrac{dQ}{dt} = \dfrac{VC}{2} \times \dfrac{2}{3RC} e^{-2t/3RC} = \dfrac{V}{3R} e^{-2t/3RC}$



16. Apply KVL in EBCDE

$-iR - \dfrac{dQ}{dt}R - \dfrac{Q}{C} = 0 \Rightarrow i = -\dfrac{dQ}{dt} + \dfrac{Q}{RC}$

$\Rightarrow \quad i = \left[\dfrac{V}{3R} e^{-2t/3RL} + \dfrac{V}{2R}(1 - e^{-2t/3RC})\right]$

$\Rightarrow \quad i = -\dfrac{V}{R}\left[\dfrac{1}{2} - \dfrac{e^{-2t/3RC}}{6}\right] = \dfrac{V}{6R}(e^{-2t/3RC} - 3)$

SECTION –C

2. There is no potential difference across the plates of capacitor so charge = 0

3. Use the expressions, $S = R_1 + R_2$ & $P^{-1} = R_1^{-1} + R_2^{-1}$

4. Particle will cross the square if it crosses the point of maximum potential between A and C–which is C itself.

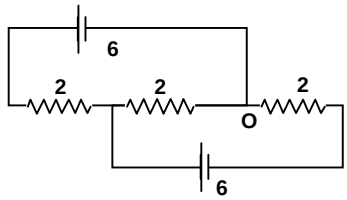
$\dfrac{1}{2}mv_0^2 = K + \dfrac{1}{4\pi\varepsilon_0}\dfrac{4Q^2q}{(l/\sqrt{2})}$

$K > 0 \Rightarrow V_0^2 = 8\sqrt{2}\left(\dfrac{1}{4\pi\varepsilon_0}\right)\left(\dfrac{Qq}{ml}\right)$

5. $\frac{x+6-6}{2}+\frac{x+6-0}{2}+\frac{x-0}{2}=0$

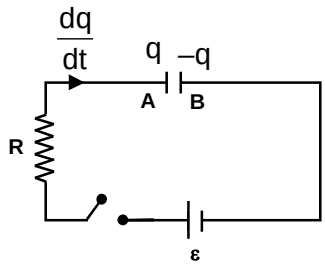
$\Rightarrow \quad x=-2$

$\therefore \quad i=\frac{x+6-0}{2}=\frac{4}{2}=2$



6. $-\frac{q}{C}+\varepsilon-\frac{dq}{dt}R=0$

$\Rightarrow \quad \int_{-Q_0/2}^{Q_0/2}\frac{dq}{q-C\varepsilon}=\int_0^t-\frac{dt}{RC}$



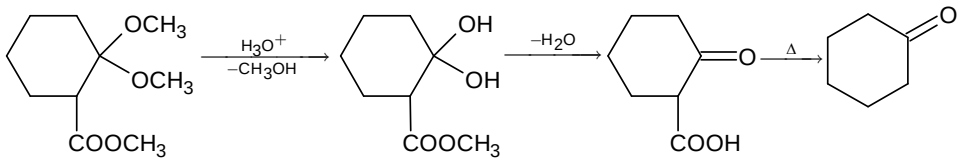
7. Use Kirchhoff law to find current in external resistance & maximise power.

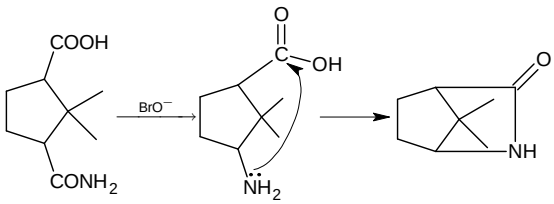
Chemistry

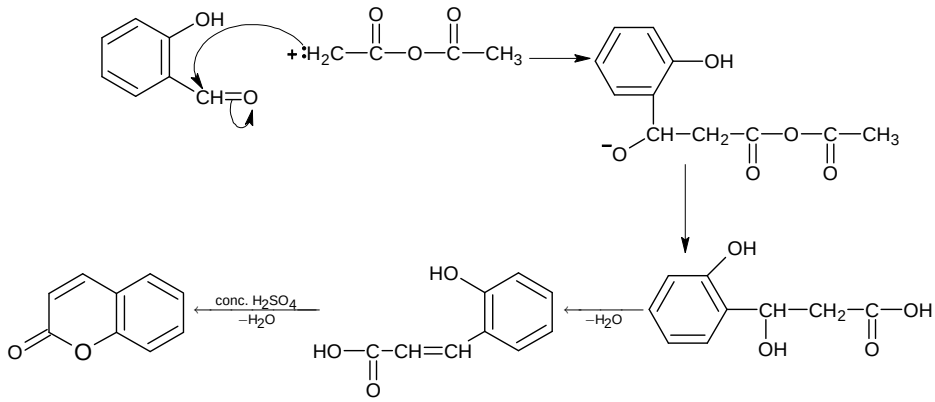
PART – II

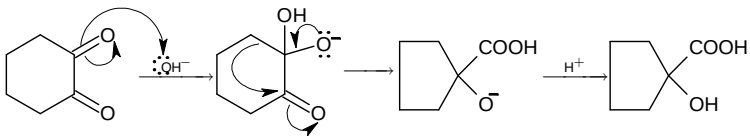
SECTION – A

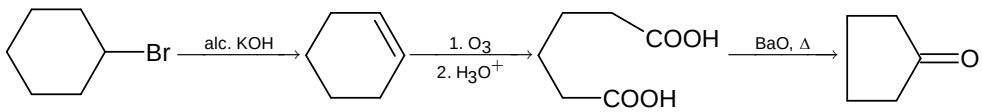
1. MnO_2 oxidized allylic alcohol.

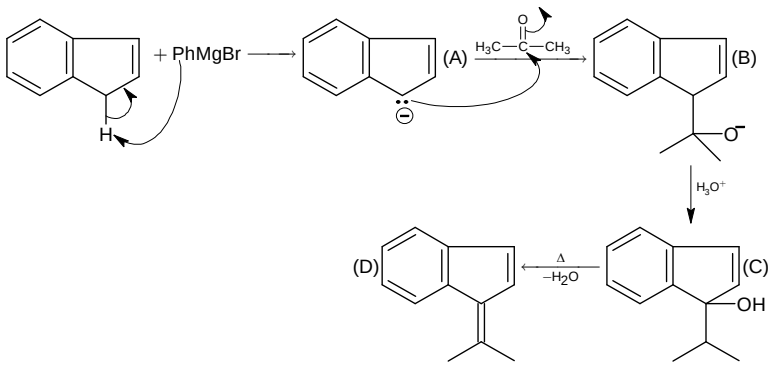
2. 

3. 

4. 

5. 

6. 

7. 

11. $NaBH_4$ does not reduce ester.

Mathematics

PART – III

SECTION – A

1. Clearly $f(x + \pi) = f(x)$, $g(x + \pi) = g(x)$ and $\phi\left(x + \frac{\pi}{2}\right) = \{(-1) f(x)\}\{(-1) g(x)\} = \phi(x)$.

2. $\int_0^{\pi/2} \sin x \log(\sin x)\,dx$

$I = \frac{1}{2}\int_0^{\pi/2} \sin x \log(\sin x)^2\,dx = \frac{1}{2}\int_0^{\pi/2} \sin x \log(1-\cos^2 x)\,dx$

$= \frac{1}{2}\int_0^1 \log(1-t^2)\,dt$, where $t = \cos x$

$= \frac{1}{2}\int_0^1 \left[-t^2 - \frac{(-t^2)^2}{2} - \frac{(-t^2)^3}{3} - \ldots\right] dt = -\left[\frac{1}{2.3} + \frac{1}{4.5} + \frac{1}{6.7} + \ldots\right] = \log_e 2 - 1 = \log_e\left(\frac{2}{e}\right)$.

3. $I = \int (f''(x)F(x) - f(x)F''(x))\,dx = \int \{(f''(x)F(x) + f'(x)F'(x)) - (f(x)F''(x) + f'(x)F'(x))\}\,dx$

$= \int d(f'(x)F(x) - f(x)F'(x)) = f'(x)\,F(x) - f(x)\,F'(x) + c$

4. $f(1) = a + b + c$, $f(-1) = a - b + c$, $f(0) = c$

$\Rightarrow a = \frac{1}{2}(f(-1) + f(1) - 2f(0))$, $b = \frac{1}{2}(f(1) - f(-1))$

$\Rightarrow f(x) = \left(\frac{1}{2}(f(-1) + f(1) - 2f(0))\right)x^2 + \left(\frac{1}{2}(f(1) - f(-1))\right)x + f(0)$

$= \frac{x(x-1)}{2} f(-1) - (x^2 - 1) f(0) + \frac{x(x+1)}{2} f(1)$

$\Rightarrow |f(x)| = \left|\frac{x(x-1)}{2} f(-1) + (1 - x^2) f(0) + \frac{x(x+1)}{2} f(1)\right|$

$2|f(x)| \le |x(x + 1)| + 2|1 - x^2| + |x(x - 1)|$ $\quad \because |f(x)| \le 1;\ x \in [-1, 1]$

$\Rightarrow$ For $x \in [1, \infty] \Rightarrow 2|f(x)| \le x(x + 1) + 2(x^2 - 1) + x(x - 1)$

$\Rightarrow |f(x)| \le 2x^2 - 1\ \forall\ x \in [1, \infty)$

$\Rightarrow g(x) \le 2x^2 - 1$

$\Rightarrow \int_1^2 g(x)\,dx \le \int_1^2 (2x^2 - 1)\,dx = \frac{11}{3}$.

5. Let $h(x) = f(x) - \sin x$

$\Rightarrow$ According to Rolle's theorem for $c \in \left(-\frac{\pi}{2}, 0\right)$, $h'(c) = 0 \Rightarrow f'(c) = \cos c$

and $d \in \left(0, \frac{\pi}{2}\right)$, $h'(d) = 0 \Rightarrow f'(d) = \cos d$

let $h'(x) = f'(x) - \cos x$, $x \in [c, d]$.

According to Rolle's theorem $h''(x) = 0$, $x \in (c, d)$

$\Rightarrow f''(x) + \sin x = 0, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\Rightarrow |f''(x)| = |\sin x| < 1$ for atleast one $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

6. $I = \int \frac{\frac{1}{x^4} - \frac{1}{x^7}}{\sqrt{\frac{2}{x^6} - \frac{4}{x^3} + 6}} dx$

Let $\frac{2}{x^6} - \frac{4}{x^3} + 6 = t^2$

$\Rightarrow I = \frac{1}{6}\int dt = \frac{1}{6}\sqrt{\frac{2}{x^6} - \frac{4}{x^3} + 6} + c = \frac{\sqrt{6x^6 - 4x^3 + 2}}{6x^3} + c$.

7. $\int_0^{\pi/6} 0dx + \int_{\pi/6}^{\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)} 1dx + \int_{\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)}^{\pi/3} 2dx = \frac{\pi}{2} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right) = \cot^{-1}\left(\frac{2}{\sqrt{3}}\right)$

8. $\lim_{x\to -4} \frac{\tan \pi x}{x+4} \quad \left(\frac{0}{0}\right)$

$= \lim_{x\to -4} \frac{\pi \sec^2 \pi x}{1} = \pi$

$\lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^x = \lim_{x\to\infty} e^{x\cdot\frac{1}{x^2}} = 1$

given limit = $\pi + 1$.

9. Since, according to option (A), we have
sin [x] = cos [x]

$\Rightarrow [x] = n\pi + \left(\frac{\pi}{4}\right) = \left(n + \frac{1}{4}\right)\pi$

Which is not possible, as LHS $\in$ I and RHS $\in Q^c$
Hence, sin [x] = cos [x] has no solution
However, sin [x] = tan [x]
$\Rightarrow [x] = n\pi$, $n \in I$
Which is possible when [x] = 0 and n = 0
$\therefore 0 \le x < 1$

10. 'g' is decreasing
$\Rightarrow g(1) \le g(0)$
Therefore f is increasing
$\Rightarrow f\{g(0)\} \ge f\{g(1)\}$
$\therefore 0 \ge f\{g(1)\}$, from the question. But, $f(x) \ge 0$, from the question
$\therefore f\{g(1)\} = 0$
As $x \ge 0$, $g(x) \le g(0)$
So, $f\{g(x)\} \le f\{g(0)\}$
As $x \ge 0$, $f(x) \ge f(0)$
So, $g\{f(x)\} \le g\{f(0)\}$

11. Since f(g(x)) is a one-one function
$\Rightarrow f(g(x_1)) \ne f(g(x_2))$ whenever $x_1 \ne x_2$

$\Rightarrow g(x_1) \neq g(x_2)$ whenever $x_1 \neq x_2$
$\Rightarrow g(x)$ must be one-one.
Let $f(x) = y$ is satisfied by $x = x_1$ and x_2
If $g(x)$ is such that its range has only one of x_1 and x_2, then $f(g(x))$ can be one-one even if $f(x)$ is many-one

12. $f(x)$ = odd degree polynomial + bounded function $\cot^{-1}x \in (0, \pi)$
also, $f'(x) > 0$.

13. $f(x) = x^4 + 1 + \dfrac{1}{x^2+x+1}$ = even degree polynomial + bounded function $\dfrac{1}{x^2+x+1} \in \left(0, \dfrac{4}{3}\right)$

$f'(x) = \dfrac{4x^3(x^2+x+1)^2 - 2x - 1}{(x^2+x+1)^2}$

$\Rightarrow f'(x) = 0$ has atleast one root which is repeated odd number of times or it has one root which is not repeated since numerator of $f'(x)$ is a polynomial of degree 7.
$\Rightarrow f(x) = 0$ has a point of extrema.

16. $P(1) \equiv (-2, -2)$

$\dfrac{dx}{dt} = -6t$, $\dfrac{dy}{dt} = 1 - 9t^2$

$\Rightarrow m_t = \dfrac{9t^2 - 1}{6t}$

at $t = 1$, $m_1 = \dfrac{4}{3}$

$\Rightarrow$ equation of tangent at P

$y + 2 = -\dfrac{4}{3}(x + 2) \Rightarrow (t + 1)^2 (3t - 2) = 0 \Rightarrow t = \dfrac{2}{3}$

$\Rightarrow Q \equiv \left(-\dfrac{4}{3}, -\dfrac{2}{9}\right)$

SECTION – C

1. $\displaystyle\int_0^2 7\left\{\frac{x^2(x^4-4x-3)}{(x^3-1)^2}\cos x - \frac{(x^4+1)}{(x^3-1)}\sin x\right\}dx$

Substitute $y = \dfrac{x^4+1}{x^3-1}\cos x$

$\displaystyle 7\int_0^2 dy = y\Big|_0^2 = \frac{24}{7} \times 7 = 24.$

2. $\displaystyle I = \int_0^1 0\,dx + \int_1^{\sqrt{2}} dx + \int_{\sqrt{2}}^{1.5} 2\,dx = (\sqrt{2}-1) + 2(1.5 - \sqrt{2}) = 2 - \sqrt{2}$

$[I] = [0.41] = 0$

3. Let $y = \dfrac{3x^2+2x-1}{x^2+x+1}$

$\Rightarrow (y - 3)x^2 + (y - 2)x + y + 1 = 0.$
As $x \in R$, $D \geq 0 \Rightarrow (y - 2)^2 - 4(y - 3)(y + 1) \geq 0$
$\Rightarrow 3y^2 - 4y - 16 \leq 0$
Clearly y lies between the roots.

Roots of the equation $3y^2 - 4y - 16 = 0$ are

$y = \frac{4 \pm \sqrt{208}}{6} = \frac{2 \pm \sqrt{52}}{3}$

$\therefore$ Absolute maximum of the expression = $\frac{2+\sqrt{52}}{3} = \alpha$

$\Rightarrow [\alpha] = 3$

4. $\int_0^1 x \log x \, dx = \int_0^1 \log x . x \, dx = \left| \log x . \frac{x^2}{2} \right|_0^1 - \int_0^1 \frac{1}{x} . \frac{x^2}{2} dx = -\frac{1}{4}$

5. Let $p(x) = 3 \sin \sqrt{\frac{\pi^2}{9} - x^2}$ and $g(x) = \left| |x| - \frac{3\sqrt{3}}{2} \right|$.

$\Rightarrow$ p(x) and g(x) are even functions

and p(x) decreases in $x \in \left[0, \frac{\pi}{3}\right]$

and g(x) decreases in $x \in \left[0, \frac{3\sqrt{3}}{2}\right]$.

Hence p(x) and g(x) will be maximum at x = 0

$\Rightarrow$ maximum value of f(x) is $f(0) = 3\sqrt{3}$.

6. $f(x) = \begin{cases} 2; & x = 0 \\ -2; & x \in R - 0 \end{cases}$ $f(x) = \begin{cases} -2; & x = 0 \\ 2; & x \in R - 0 \end{cases}$

7. Since f(x) and g(x) are one-one and onto and are also the mirror images of each other with respect to the line y = 2. It clearly indicates that h(x) = f(x) + g(x) will be a constant function and will always be equal to 4.